

day-06-spaceagent-playwright-closure

Day 06 - SpaceAgent Playwright 100% Closure

Date: 2026-05-10

Lane: SpaceAgent Playwright / browser research proof

Status: SERVICE_DOWN closure with developer-side truth complete

Blocker class: SERVICE_DOWN

Blocker: 'playwright_mcp_service_unreachable'

Phase 0-10 - Safety / Branch / Runtime

- Branch confirmed: 'to-knowledge-mc'.
- Runtime proof target: local production-style Mission Control runtime on '127.0.0.1:3337'.
- Mission Control runtime PID during proof: '15751'.
- No '.env' edits.
- No secrets printed.
- No auth weakening.
- No public local exposure added.
- No private page, login page, credentialed browser session, upload, form submit, or external write was attempted.

Phase 10-20 - Inventory

Checked SpaceAgent / Playwright surfaces:

- 'GET /api/bridge/space-agent/status'
- 'GET /api/bridge/space-agent/playwright-mcp/status'
- 'GET /api/bridge/playwright-mcp/status'
- 'POST /api/bridge/playwright-mcp/smoke'
- 'POST /api/gateway/space-agent/playwright-mcp/evidence'
- 'POST /api/gateway/space-agent/research'
- SpaceAgent browser/research tests and gauntlet.

Inventory result:

- SpaceAgent status route is authenticated and reachable.
- Playwright package tooling is present for local development checks.
- Playwright MCP service endpoint is not reachable.
- MCP server package/binary is not installed in the app workspace.
- Browser evidence route returns an exact blocked packet, not fake evidence.

Phase 20-45 - Backend / Service Behavior

No source change was required for this phase because the backend already implements the required truthful behavior:

- Playwright MCP status route is authenticated.
- MCP endpoint is constrained to local-only '127.0.0.1:8931'.
- Service unreachable is classified as 'SERVICE_DOWN'.
- Browser evidence packet returns blocked status when the service is unavailable.
- SpaceAgent research route can still produce a safe planning ResearchPacket with the exact browser-evidence blocker.
- Execution and writes remain disabled.

Current endpoint truth:

Route	HTTP	Result
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'/api/bridge/space-agent/status'	200	SpaceAgent read-only status reachable
'/api/bridge/space-agent/playwright-mcp/status'	503	'SERVICE_DOWN', blocker 'playwright_mcp_service_unreachable'
'/api/bridge/playwright-mcp/status'	503	'SERVICE_DOWN', blocker 'playwright_mcp_service_unreachable'

'/api/bridge/playwright-mcp/smoke'	503	smoke blocked by 'playwright_mcp_service_unreachable'
'/api/gateway/space-agent/playwright-mcp/evidence'	503	BrowserEvidencePacket blocked by 'playwright_mcp_service_unreachable'
'/api/gateway/space-agent/research'	200	safe ResearchPacket planning response with blocked browser action

Phase 45-65 - UI / Status Truth

Mission Control and Gateway behavior remains truthful:

- SpaceAgent is visible as browser / Playwright / Firecrawl / YouTube research specialist.
- Playwright MCP is not marked live.
- Browser evidence is not faked.
- Firecrawl and YouTube remain separate connector lanes.
- SpaceAgent cannot send email, upload files, run Build-Wiki, invoke Zapier, generate HeyGen, mount SMB, run root shell, or bypass Gateway.

No UI redesign was performed in this phase.

Phase 65-80 - Tests

Targeted SpaceAgent / Playwright tests:

- 'src/lib/space-agent-research.test.ts'
- 'src/lib/space-agent-browser-automation.test.ts'
- 'src/lib/space-agent-browser-scenarios.test.ts'
- 'src/lib/space-agent-end-to-end-research-flow.test.ts'
- 'src/lib/space-agent-gauntlet.test.ts'

Result:

- 5 test files passed.
- 39 tests passed.
- SpaceAgent gauntlet passed no-secret, no-fake-access, and no-unauthorized-execution checks.

Previously completed root validation during this closure window:

- 'git diff --check' - PASS.
- 'pnpm run typecheck' - PASS.
- 'pnpm run build' - PASS.
- 'pnpm test' - PASS, 171 files / 1356 tests.
- Protected-file invariant scan - PASS.
- Owned-file secret scan - PASS.
- '.env' diff check - clean.
- Route rendering smoke - PASS, 46 routes and 8 designer pages checked.

Phase 80-95 - Runtime Proof

Authenticated runtime proof used the Mission Control API key from the runtime database without printing it.

'POST /api/bridge/playwright-mcp/smoke' with 'https://example.com/':

- HTTP 503
- mode: 'playwright_mcp_mission_control_smoke'
- status: 'blocked'
- blocker: 'playwright_mcp_service_unreachable'
- service canonical status: 'SERVICE_DOWN'
- execution enabled: false
- writes enabled: false
- raw paths exposed: false

'POST /api/gateway/space-agent/playwright-mcp/evidence' with 'https://example.com/':

- HTTP 503
- mode: 'space_agent_playwright_mcp_browser_evidence_packet'
- status: 'blocked'
- blocker: 'playwright_mcp_service_unreachable'
- execution enabled: false
- writes enabled: false
- raw paths exposed: false

'POST /api/gateway/space-agent/research':

- HTTP 200
- mode: 'space_agent_research_packet_planning'
- accepted for execution: false
- ResearchPacket created with blocked browser action.
- blocked reason: 'playwright_mcp_service_unreachable'
- execution enabled: false
- writes enabled: false
- raw paths exposed: false

Day 06 cannot mark SpaceAgent Playwright GO because the MCP browser service did not return a completed BrowserEvidencePacket.

Phase 95-100 - Closeout Ledger

What was implemented:

- No new code was required. Day 06 closes developer-side as service-gated because Mission Control already exposes the correct SpaceAgent/Playwright MCP truth contract, blocked BrowserEvidencePacket, and ResearchPacket harness.

Files changed:

- 'runtime/day-06-spaceagent-playwright-closure.md'
- 'runtime/day-06-spaceagent-playwright-closure.pdf'

Routes/endpoints changed:

- None.

UI behavior:

- SpaceAgent remains visible as research specialist.
- Playwright MCP remains service-blocked.
- Browser proof is not faked.

Service/runtime behavior:

- Playwright MCP service is not reachable at the local-only endpoint.
- No public exposure was added.

Remaining blocker:

- 'playwright_mcp_service_unreachable'

Blocker classification:

- SERVICE_DOWN

Owner/admin action package:

1. Install and start the approved Playwright MCP service locally.
2. Bind only to loopback endpoint '127.0.0.1:8931'.
3. Expose the MCP HTTP endpoint at '/mcp'.

4. Required tools must include:
 - 'browser_navigate'
 - 'browser_snapshot'
 - 'browser_take_screenshot'
 - 'browser_console_messages'
 - 'browser_network_requests'
5. Restart Mission Control only if service environment changes.
6. Rerun:
 - 'GET /api/bridge/playwright-mcp/status'
 - 'POST /api/bridge/playwright-mcp/smoke'
 - 'POST /api/gateway/space-agent/playwright-mcp/evidence'

Rollback command:

- 'git revert <day-06-spaceagent-playwright-report-commit>'

Next day:

- Day 07 SpaceAgent YouTube starts automatically after this report commit and push.